

ADVANCED STATISTICS PROJECT REPORT

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PROBLEM STATEMENT 1A

Salary is hypothesized to depend on educational qualification and occupation. To understand the dependency, the salaries of 40 individuals [[SalaryData.csv](#)] are collected and each person's educational qualification and occupation are noted. Educational qualification is at three levels, High school graduate, Bachelor, and Doctorate. Occupation is at four levels, Administrative and clerical, Sales, Professional or specialty, and Executive or managerial. A different number of observations are in each level of education – occupation combination.

DATA ATTRIBUTES

1. Education : Educational qualification (high school graduate, bachelor, doctorate,etc)
2. Occupation : occupation of different people (administrative and clerical, professional,etc)
3. Salary : continuous from 50103 to 260150.

SAMPLE OF THE DATASET

	Education	Occupation	Salary
0	Doctorate	Adm-clerical	153197
1	Doctorate	Adm-clerical	115945
2	Doctorate	Adm-clerical	175935
3	Doctorate	Adm-clerical	220754
4	Doctorate	Sales	170769

Table 1. Dataset sample (problem 1)

Dataset has 3 variables with 3 different educational qualification and 4 different occupation. Depending upon the educational and occupation salary of the people is defined.

[Check for the type of variables in the dataframe.](#)

Education	object
Occupation	object
Salary	int64

There are total 40 rows and 3 columns in the dataset. Out of 3, 2 columns are of object type and 1 is of integer data type.

[Check for missing values in the dataset.](#)

Education	40 non-null
Occupation	40 non-null
Salary	40 non-null

[1.1 State the null and the alternate hypothesis for conducting one-way ANOVA for both Education and Occupation individually.](#)

Hypothesis for Education :

H_0 : mean salary is same for all education levels

H_a : mean salary is different for atleast one of the education levels

Hypothesis for Occupation :

H₀: mean salary is same for every occupation

H_a: mean salary is different for atleast one of the occupation

1.2 Perform a one-way ANOVA on Salary with respect to Education. State whether the null hypothesis is accepted or rejected based on the ANOVA results.

Here, the level of significance is 0.05 and confidence interval is 0.95.

After performing one-way ANOVA on salary with respect to education the following results were obtained :

	df	sum_sq	mean_sq	F	PR(>F)
C(Education)	2.0	1.026955e+11	5.134773e+10	30.95628	1.257709e-08
Residual	37.0	6.137256e+10	1.658718e+09	NaN	NaN

Table 2. ANOVA on salary w.r.t education

Here, pvalue is 1.257709e-08 which is smaller than the level of significance i.e 0.05.

Thus, null hypothesis is rejected based on the above observation and it is concluded that mean salary is not same for all the educational levels.

1.3 Perform a one-way ANOVA on Salary with respect to Occupation. State whether the null hypothesis is accepted or rejected based on the ANOVA results.

Here, the level of significance is 0.05 and confidence interval is 0.95.

After performing one-way ANOVA on salary with respect to Occupation the following results were obtained :

	df	sum_sq	mean_sq	F	PR(>F)
C(Occupation)	3.0	1.125878e+10	3.752928e+09	0.884144	0.458508
Residual	36.0	1.528092e+11	4.244701e+09	NaN	NaN

Table 3. ANOVA on salary w.r.t occupation

Here, pvalue is 0.458508 which is greater than the level of significance i.e 0.05. Thus, we fail to reject the null hypothesis based on the above the observation and it is concluded that mean salary is same for all the occupations.

1.4 If the null hypothesis is rejected in either (2) or in (3), find out which class means are significantly different. Interpret the result.

Null hypothesis is rejected in (2) thus, mean salary is different for atleast one of the education levels i.e. salary of bachelor, high school graduate or doctorate is not same.

PROBLEM 1B

1.5 What is the interaction between two treatments? Analyze the effects of one variable on the other (Education and Occupation) with the help of an interaction plot.

Interaction effects represent the combined effects of factors on the dependent measure. When an interaction effect is present, the impact of one factor depends on the level of the other factor.

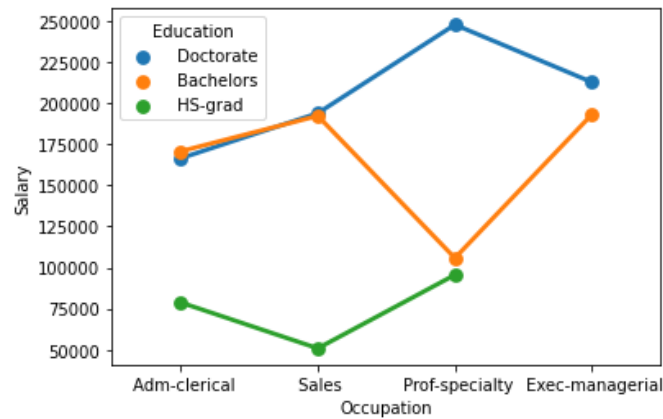


Fig 1. Interaction plot

We can see that there is some sort of interaction between the two treatments. So, we will introduce a new term while performing the two-way ANOVA.

1.6 Perform a two-way ANOVA based on Salary with respect to both Education and Occupation (along with their interaction Education*Occupation). State the null and alternative hypotheses and state your results. How will you interpret this result?

The null and alternate hypothesis for two-way ANOVA is given as :

H₀: There is no interaction effect of education and occupation on salary

H_a: There is interaction effect of education and occupation on salary

After performing two-way ANOVA based on salary with respect to both Education and Occupation (along with their interaction) the following results are obtained :

	df	sum_sq	mean_sq	F	\
C(Education)	2.0	1.026955e+11	5.134773e+10	72.211958	
C(Occupation)	3.0	5.519946e+09	1.839982e+09	2.587626	
C(Education):C(Occupation)	6.0	3.634909e+10	6.058182e+09	8.519815	
Residual	29.0	2.062102e+10	7.110697e+08	NaN	
					PR(>F)
C(Education)					5.466264e-12
C(Occupation)					7.211580e-02
C(Education):C(Occupation)					2.232500e-05
Residual					NaN

Table 4. Two-way ANOVA with interaction

Here, the pvalue is **2.232500e-05** and level of significance is 0.05.

Thus, for a given level of significance the null hypothesis is rejected as pvalue is less than 0.05. Therefore, salary is affected by both education and occupation simultaneously.

1.7 Explain the business implications of performing ANOVA for this particular case study.

ANOVA checks the impact of one or more factors by comparing the means of different samples. In the above case study ANOVA helps us to compare the effect of education and occupation on salary of an individual.

We concluded that :

1. Salary is affected by different educational levels. A high school graduate, bachelor and a doctorate will get different salaries as per their qualification.

- Salary is not affected by different occupations but together occupation and education affects salary of an individual as they have some interaction present.

PROBLEM STATEMENT 2

The dataset [Education - Post 12th Standard.csv](#) contains information on various colleges. You are expected to do a Principal Component Analysis for this case study according to the instructions given. The data dictionary of the 'Education - Post 12th Standard.csv' can be found in the following file: [Data Dictionary.xlsx](#).

DATA ATTRIBUTES

Names: Names of various university and colleges

Apps: Number of applications received

Accept: Number of applications accepted

Enroll: Number of new students enrolled

Top10perc: Percentage of new students from top 10% of Higher Secondary class

Top25perc: Percentage of new students from top 25% of Higher Secondary class

F.Undergrad: Number of full-time undergraduate students

P.Undergrad: Number of part-time undergraduate students

Outstate: Number of students for whom the particular college or university is Out-of-state.

Room.Board: Cost of Room and board

Books: Estimated book costs for a student

Personal: Estimated personal spending for a student

PhD: Percentage of faculties with Ph.D.'s

Terminal: Percentage of faculties with terminal degree

S.F.Ratio: Student/faculty ratio

perc.alumni: Percentage of alumni who donate

Expend: The Instructional expenditure per student

Grad.Rate: Graduation rate

SAMPLE OF THE DATASET

Names	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad.Rate
Abilene Christian University	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	18.1	12	7041	60
Adelphi University	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	12.2	16	10527	56
Adrian College	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	12.9	30	8735	54
Agnes Scott College	417	349	137	60	89	510	63	12960	5450	450	875	92	97	7.7	37	19016	59
Alaska Pacific University	193	146	55	16	44	249	869	7560	4120	800	1500	76	72	11.9	2	10922	15

Table 5. Dataset sample (problem 2)

Dataset has 18 variables with 777 different universities. Each university has different number of applications received, different acceptance rate, etc.

[Check for the type of variables in the dataframe](#)

```
Names          object
Apps           int64
Accept         int64
Enroll         int64
Top10perc      int64
Top25perc      int64
```

F.Undergrad	int64
P.Undergrad	int64
Outstate	int64
Room.Board	int64
Books	int64
Personal	int64
PhD	int64
Terminal	int64
S.F.Ratio	float64
perc.alumni	int64
Expend	int64
Grad.Rate	int64

There are total 777 rows and 18 columns in the dataset. Out of 18, 1 column is of object data type and 1 is of float data type, rest other are of integer data type.

[Check for missing values in the dataset.](#)

Names	777 non-null
Apps	777 non-null
Accept	777 non-null
Enroll	777 non-null
Top10perc	777 non-null
Top25perc	777 non-null
F.Undergrad	777 non-null
P.Undergrad	777 non-null
Outstate	777 non-null
Room.Board	777 non-null
Books	777 non-null
Personal	777 non-null
PhD	777 non-null
Terminal	777 non-null
S.F.Ratio	777 non-null
perc.alumni	777 non-null
Expend	777 non-null
Grad.Rate	777 non-null

[2.1 Perform Exploratory Data Analysis \[both univariate and multivariate analysis to be performed\].](#)

[What insight do you draw from the EDA?](#)

Univariate analysis of each variable is as follows :

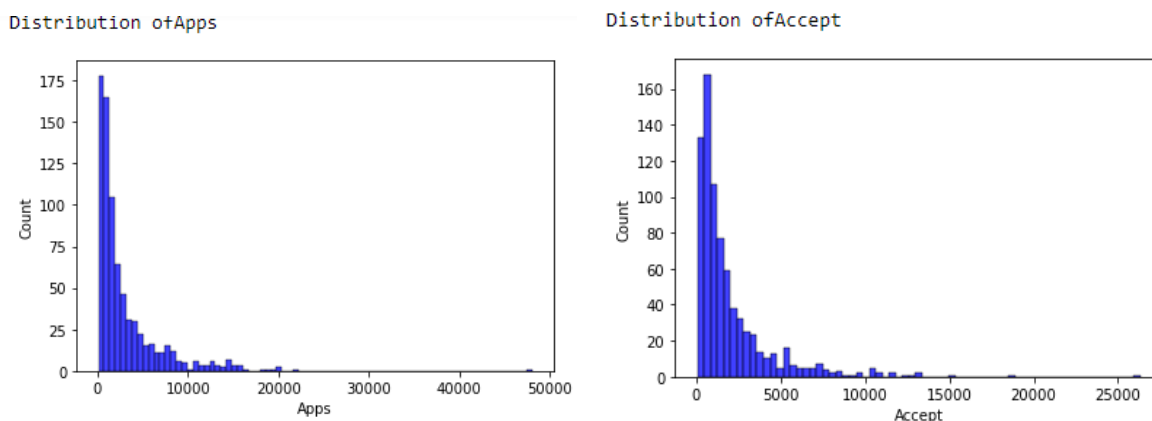


Fig 2. Distribution of apps and accept

- Distribution of apps and accept is right skewed.

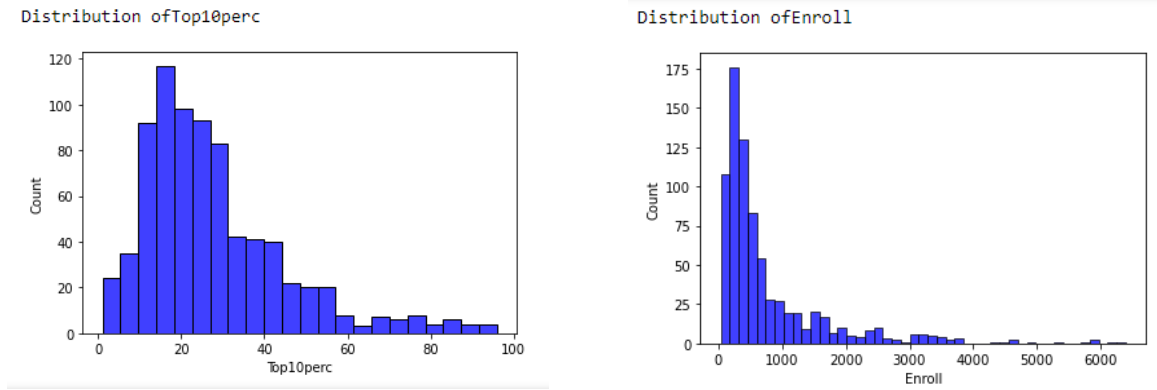


Fig 3. Distribution of top 10 perc and enroll

- Distribution of top 10 perc and enrol is also right skewed.

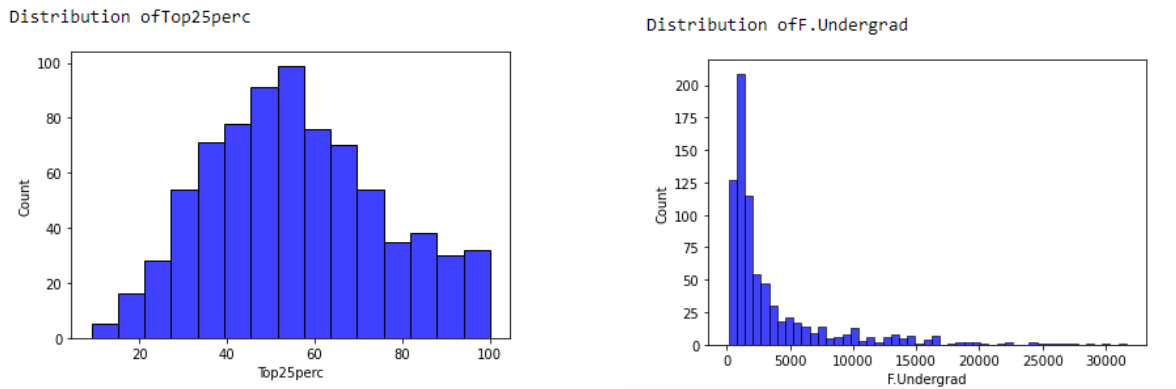


Fig 4. Distribution of top 25 perc and F.undergrad

- Distribution of top 25 perc looks like approximately normal while F.undergrad also has skewed data points.

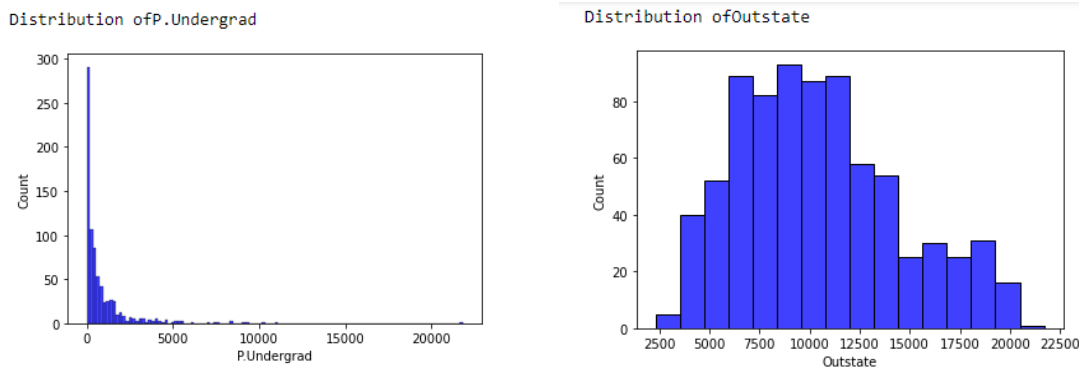
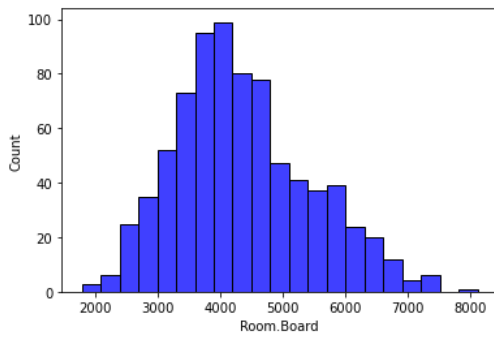


Fig 5. Distribution of P.undergrad and outstate

- Distribution of P.undergrad is also skewed towards right while offstate also is skewed.

Distribution ofRoom.Board



Distribution ofBooks

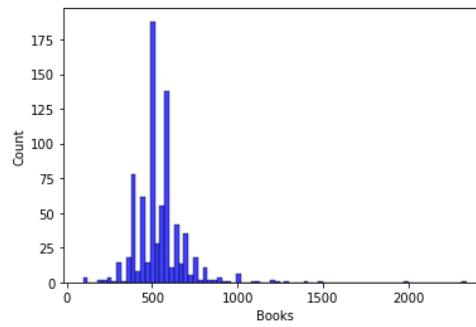
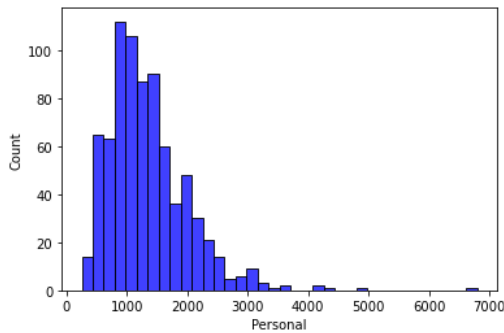


Fig 6. Distribution of room board and books

- Distribution of room board looks approximately normal while books has unequal distribution.

Distribution ofPersonal



Distribution ofPhD

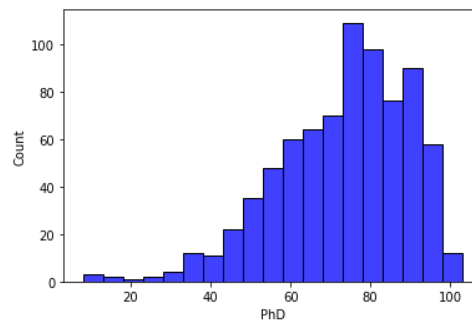
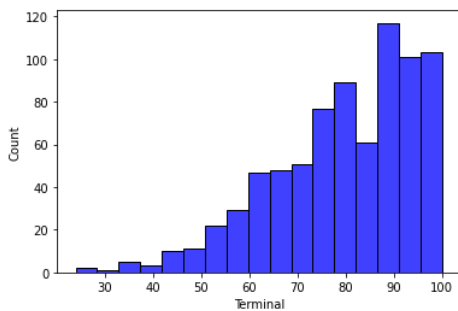


Fig 7. Distribution of personal and PhD

- Distribution of personal is right skewed while that of PhD is left skewed.

Distribution ofTerminal



Distribution ofS.F.Ratio

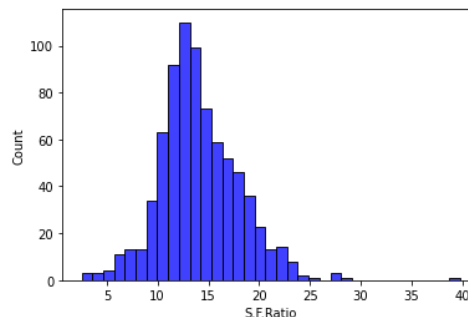
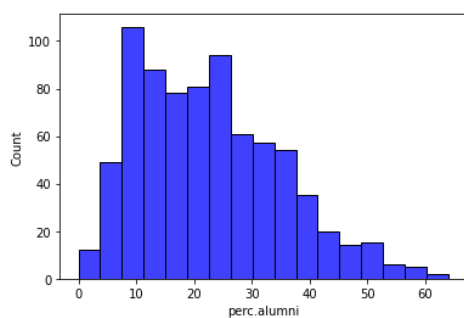


Fig 8. Distribution of terminal and S.F. ratio

- Distribution of terminal is left skewed while of S.F ratio looks to be approximately normal.

Distribution ofperc.alumni



Distribution ofExpend

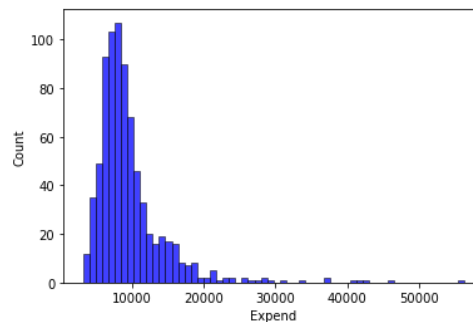


Fig 9. Distribution of perc alumni and expend

- Distribution of perc alumni is unequal while that of expend is right skewed.

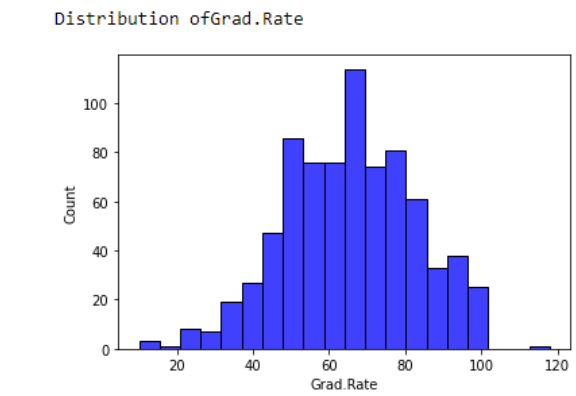


Fig 10. Distribution of grad rate

- Unequal distribution persists in grad rate.

Multivariate analysis of the dataset is as follows :

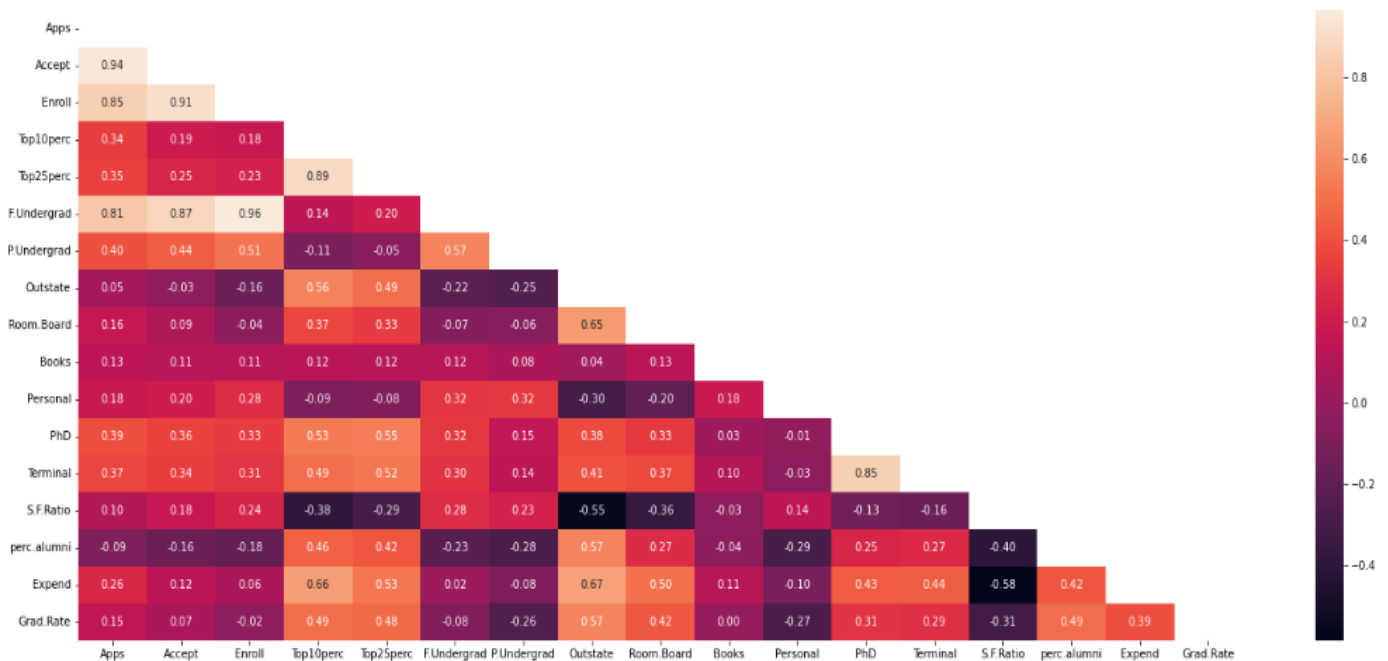


Fig 11. Correlation heatmap

From the correlation plot, we can see that accept and apps are highly correlated. Also, there is 0.96 correlation between F.undergrad and enrol. Correlation values near to 1 or -1 are highly positively correlated and highly negatively correlated respectively.

2.2 Is scaling necessary for PCA in this case? Give justification and perform scaling.

Yes, scaling is necessary for PCA in this case as we have to compare the data. Few values are in thousands and few values are only in two digits so without scaling they won't give the accurate results.

After performing scaling, the following dataset is obtained.

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio	perc.alumni	Expend	Grad.Rate
0	-0.346882	-0.321205	-0.063509	-0.258583	-0.191827	-0.168116	-0.209207	-0.746356	-0.964905	-0.602312	1.270045	-0.163028	-0.115729	1.013776	-0.867574	-0.501910	-0.318252
1	-0.210884	-0.038703	-0.288584	-0.655656	-1.353911	-0.209788	0.244307	0.457496	1.909208	1.215880	0.235515	-2.675646	-3.378176	-0.477704	-0.544572	0.166110	-0.551262
2	-0.406866	-0.376318	-0.478121	-0.315307	-0.292878	-0.549565	-0.497090	0.201305	-0.554317	-0.905344	-0.259582	-1.204845	-0.931341	-0.300749	0.585935	-0.177290	-0.687767
3	-0.668261	-0.681682	-0.692427	1.840231	1.677612	-0.658079	-0.520752	0.626633	0.996791	-0.602312	-0.688173	1.185206	1.175657	-1.615274	1.151188	1.792851	-0.376504
4	-0.726176	-0.764555	-0.780735	-0.655656	-0.596031	-0.711924	0.009005	-0.716508	-0.216723	1.518912	0.235515	0.204672	-0.523535	-0.553542	-1.675079	0.241803	-2.939613

Table 6. scaled dataset

2.3 Comment on the comparison between the covariance and the correlation matrices from this data [on scaled data].

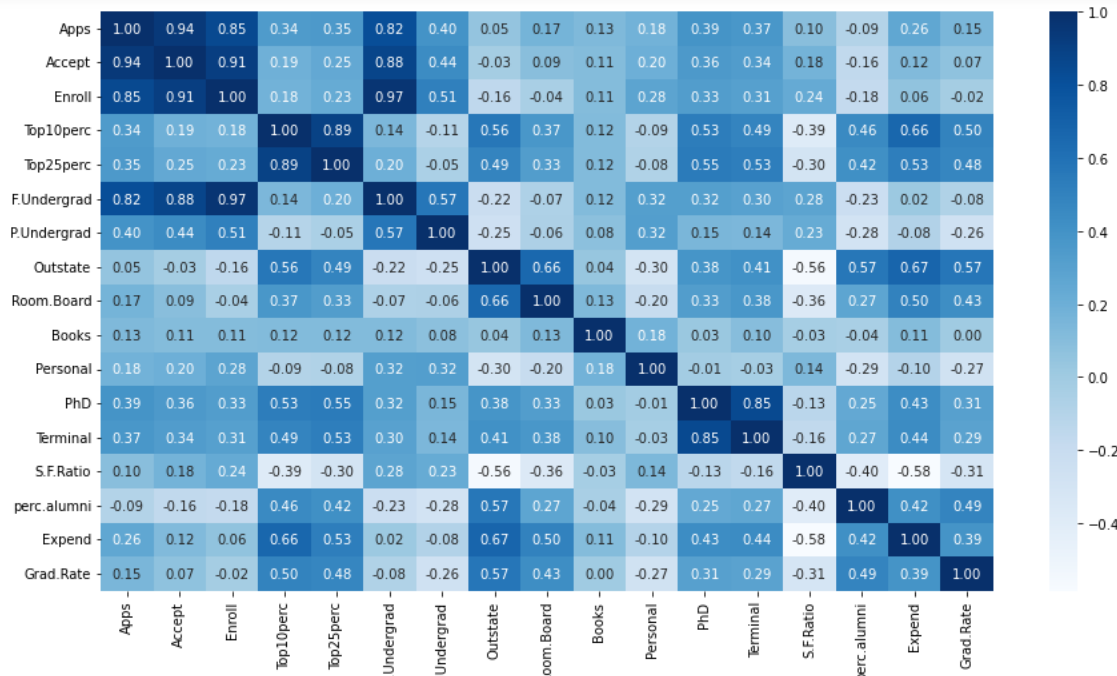


Fig 12. Covariance matrix

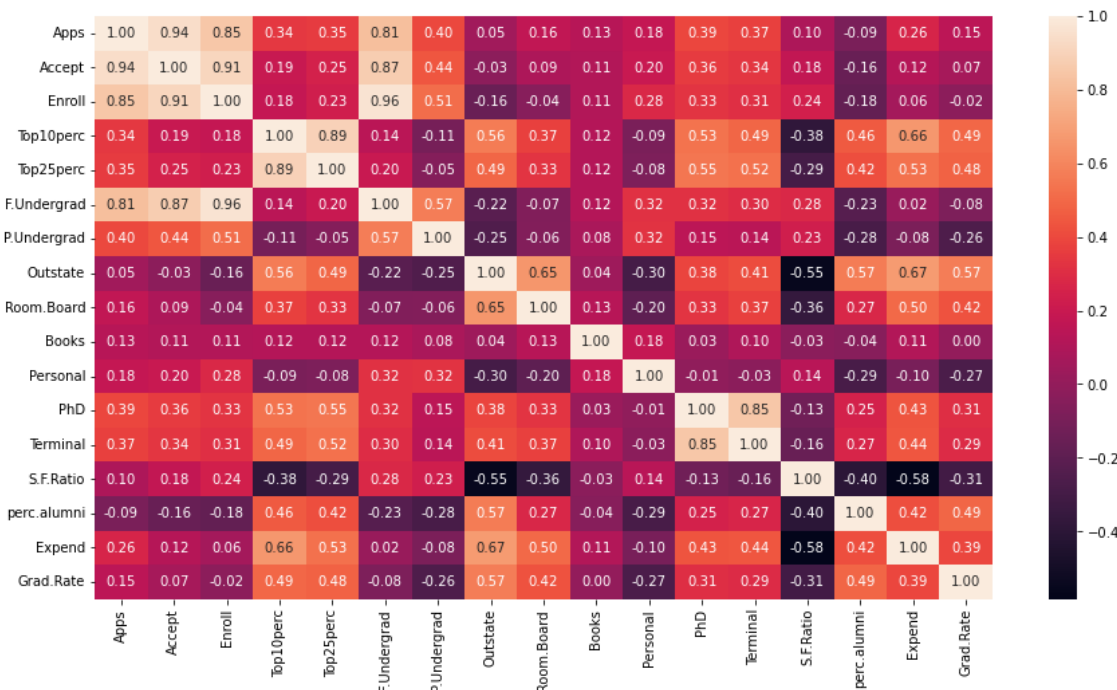


Fig 13. Correlation matrix

Here, both covariance and correlation matrix shows the same result of the scaled data.

2.4 Check the dataset for outliers before and after scaling. What insight do you derive here?

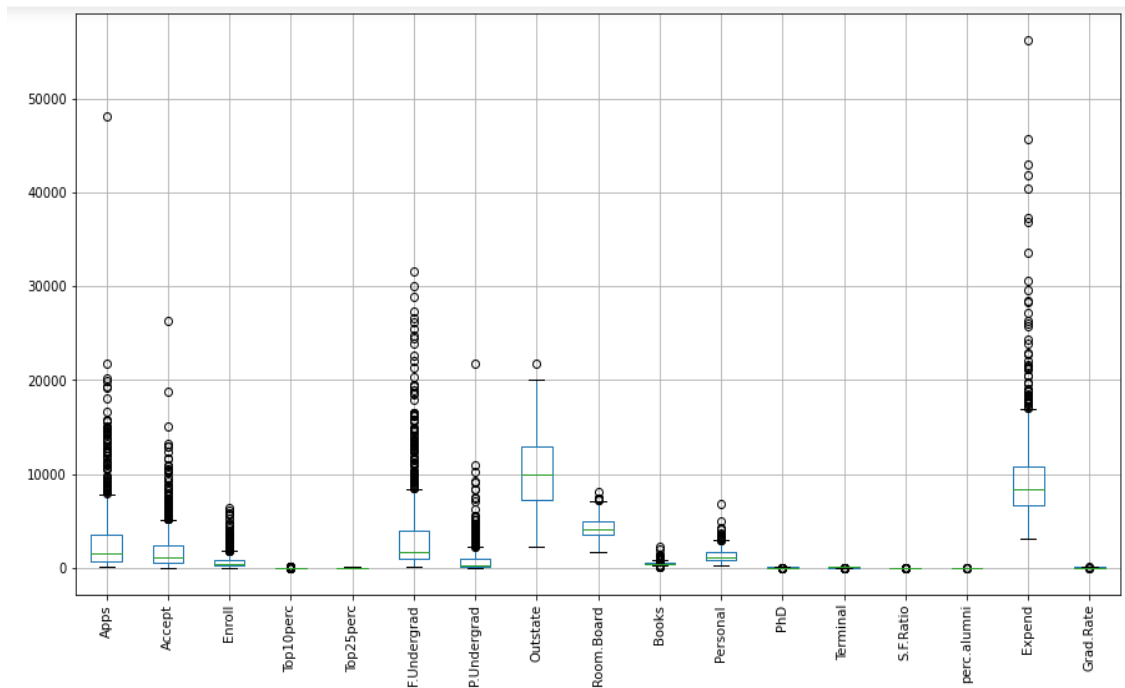


Fig 14. Boxplots before scaling

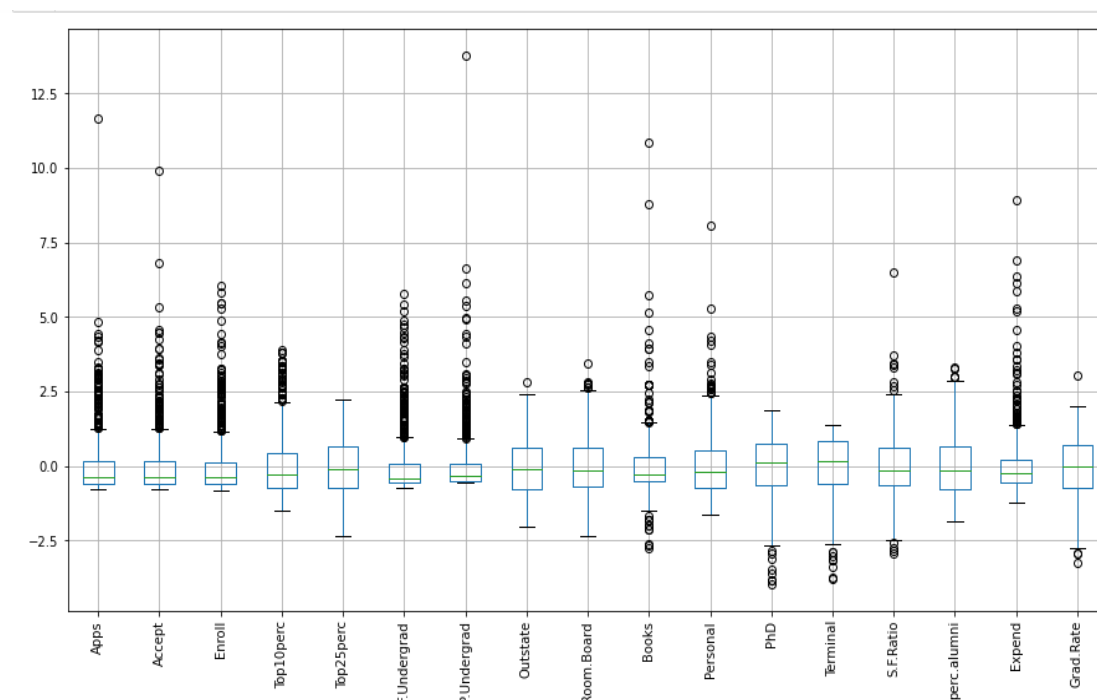


Fig 15. Boxplots after scaling

- Even after scaling the data the outliers are still present in the dataset. Boxplots helps us to detect outliers and the above figures show that outliers are present in many variables.
- Thus, scaling did not affect the outliers rather it has scaled our data by shrinking the range of the variables present.

2.5 Extract the eigenvalues and eigenvectors.

Eigen vectors are given as :

```
[ 2.48765602e-01,  2.07601502e-01,  1.76303592e-01,  
 3.54273947e-01,  3.44001279e-01,  1.54640962e-01,
```

2.64425045e-02, 2.94736419e-01, 2.49030449e-01,
 6.47575181e-02, -4.25285386e-02, 3.18312875e-01,
 3.17056016e-01, -1.76957895e-01, 2.05082369e-01,
 3.18908750e-01, 2.52315654e-01],
 [3.31598227e-01, 3.72116750e-01, 4.03724252e-01,
 -8.24118211e-02, -4.47786551e-02, 4.17673774e-01,
 3.15087830e-01, -2.49643522e-01, -1.37808883e-01,
 5.63418434e-02, 2.19929218e-01, 5.83113174e-02,
 4.64294477e-02, 2.46665277e-01, -2.46595274e-01,
 -1.31689865e-01, -1.69240532e-01],
 [-6.30921033e-02, -1.01249056e-01, -8.29855709e-02,
 3.50555339e-02, -2.41479376e-02, -6.13929764e-02,
 1.39681716e-01, 4.65988731e-02, 1.48967389e-01,
 6.77411649e-01, 4.99721120e-01, -1.27028371e-01,
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 -1.27288825e-01, -2.22311021e-01, 1.40166326e-01,
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 7.59581203e-02, -1.09267913e-01],
 [-1.62374420e-02, 7.53468452e-03, -4.25579803e-02,
 -5.26927980e-02, 3.30915896e-02, -4.34542349e-02,
 -1.91198583e-01, -3.00003910e-02, 1.62755446e-01,
 6.41054950e-01, -3.31398003e-01, 9.12555212e-02,
 1.54927646e-01, 4.87045875e-01, -4.73400144e-02,
 -2.98118619e-01, 2.16163313e-01],
 [-4.24863486e-02, -1.29497196e-02, -2.76928937e-02,
 -1.61332069e-01, -1.18485556e-01, -2.50763629e-02,
 6.10423460e-02, 1.08528966e-01, 2.09744235e-01,
 -1.49692034e-01, 6.33790064e-01, -1.09641298e-03,
 -2.84770105e-02, 2.19259358e-01, 2.43321156e-01,
 -2.26584481e-01, 5.59943937e-01],
 [-1.03090398e-01, -5.62709623e-02, 5.86623552e-02,
 -1.22678028e-01, -1.02491967e-01, 7.88896442e-02,
 5.70783816e-01, 9.84599754e-03, -2.21453442e-01,
 2.13293009e-01, -2.32660840e-01, -7.70400002e-02,
 -1.21613297e-02, -8.36048735e-02, 6.78523654e-01,
 -5.41593771e-02, -5.33553891e-03],
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 3.41099863e-01, 4.03711989e-01, -5.94419181e-02,
 5.60672902e-01, -4.57332880e-03, 2.75022548e-01,
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 1.32286331e-01, -5.90271067e-01],
 [4.30462074e-02, -5.84055850e-02, -6.93988831e-02,
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 1.00693324e-01, 1.43220673e-01, -3.59321731e-01,
 3.19400370e-02, -1.85784733e-02, 4.03723253e-02,

```

-5.89734026e-02, 4.45000727e-01, -1.30727978e-01,
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1.25997650e-01, -1.41856014e-01, -6.97485854e-02,
1.14379958e-02, 3.94547417e-02, 1.27696382e-01,
-5.83134662e-02, -1.77152700e-02, 1.04088088e-01,
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[ 8.06328039e-02, 3.34674281e-02, -8.56967180e-02,
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1.92857500e-02, -3.40115407e-02, -5.84289756e-02,
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-9.43887925e-03, -3.09001353e-03, -1.12055599e-01,
1.58909651e-01, -2.08991284e-02, -8.41789410e-03,
-2.27742017e-01, -3.39433604e-03],
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-5.27313042e-02, 1.01594830e-01, -2.59293381e-02,
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-4.38803230e-02, -5.00844705e-03],
[ 3.58970400e-01, -5.43427250e-01, 6.09651110e-01,
-1.44986329e-01, 8.03478445e-02, -4.14705279e-01,
9.01788964e-03, 5.08995918e-02, 1.14639620e-03,
7.72631963e-04, -1.11433396e-03, 1.38133366e-02,
6.20932749e-03, -2.22215182e-03, -1.91869743e-02,
-3.53098218e-02, -1.30710024e-02]

```

Eigen values are given as :

```

[5.45052162, 4.48360686, 1.17466761, 1.00820573, 0.93423123,
0.84849117, 0.6057878 , 0.58787222, 0.53061262, 0.4043029 ,
0.31344588, 0.22061096, 0.16779415, 0.1439785 , 0.08802464,
0.03672545, 0.02302787]

```

2.6 Perform PCA and export the data of the Principal Component (eigenvectors) into a data frame with the original features.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	PC12	PC13	PC14	PC15	PC16	PC17
Apps	0.248766	0.331598	-0.063092	0.281311	0.005741	-0.016237	-0.042486	-0.103090	-0.090227	0.052510	0.043046	0.024071	0.595831	0.080633	0.133406	0.459139	0.358970
Accept	0.207602	0.372117	-0.101249	0.267817	0.055786	0.007535	-0.012950	-0.056271	-0.177865	0.041140	-0.058406	-0.145102	0.292642	0.033467	-0.145498	-0.518569	-0.543427
Enroll	0.176304	0.403724	-0.082986	0.161827	-0.055694	-0.042558	-0.027693	0.058662	-0.128561	0.034488	-0.069399	0.011143	-0.444638	-0.085697	0.029590	-0.404318	0.609651
Top10perc	0.354274	-0.082412	0.035056	-0.051547	-0.395434	-0.052693	-0.161332	-0.122678	0.341100	0.064026	-0.008105	0.038554	0.001023	-0.107828	0.697723	-0.148739	-0.144986
Top25perc	0.344001	-0.044779	-0.024148	-0.109767	-0.426534	0.033092	-0.118486	-0.102492	0.403712	0.014549	-0.273128	-0.089352	0.021884	0.151742	-0.617275	0.051868	0.080348
F.Undergrad	0.154641	0.417674	-0.061393	0.100412	-0.043454	-0.043454	-0.025076	0.078890	-0.059442	0.020847	-0.081158	0.056177	-0.523622	-0.056373	0.009916	0.560363	-0.414705
P.Undergrad	0.026443	0.315088	0.139682	-0.158558	0.302385	-0.191199	0.061042	0.570784	0.560673	-0.223106	0.100693	-0.063536	0.125998	0.019286	0.020952	-0.052731	0.009018
Outstate	0.294736	-0.249644	0.046599	0.131291	0.222532	-0.030000	0.108529	0.009846	-0.004573	0.186675	0.143221	-0.823444	-0.141856	-0.034012	0.038354	0.101595	0.050900
Room.Board	0.249030	-0.137809	0.148967	0.184996	0.560919	0.162755	0.209744	-0.221453	0.275023	0.298324	-0.359322	0.354560	-0.069749	-0.058429	0.003402	-0.025929	0.001146
Books	0.064758	0.056342	0.677412	0.087089	-0.127289	0.641055	-0.149692	0.213293	-0.133663	-0.082029	0.031940	-0.028159	0.011438	-0.066849	-0.009439	0.002883	0.000773
Personal	-0.042529	0.219929	0.499721	-0.230711	-0.222311	-0.331398	0.633790	-0.232661	-0.094469	0.136028	-0.018578	-0.039264	0.039455	0.027529	-0.003090	-0.012890	-0.001114
PhD	0.318313	0.058311	-0.127028	-0.534725	0.140166	0.091256	-0.001096	-0.077040	-0.185182	-0.123452	0.040372	0.023222	0.127696	-0.691126	-0.112056	0.029808	0.013813
Terminal	0.317056	0.046429	-0.066038	-0.519443	0.204720	0.154928	-0.028477	-0.012161	-0.254938	-0.088578	-0.058973	0.016485	-0.058313	0.671009	0.158910	-0.027076	0.006209
S.F.Ratio	-0.176958	0.246665	-0.289848	-0.161189	-0.079388	0.487046	0.219259	-0.083605	0.274544	0.472045	0.445001	-0.011026	-0.017715	0.041374	-0.020899	-0.021248	-0.002222
perc.alumni	0.205082	-0.246595	-0.146989	0.017314	-0.216297	-0.047340	0.243321	0.678524	-0.255335	0.423000	-0.130728	0.182661	0.104088	-0.027154	-0.008418	0.003334	-0.019187
Expend	0.318909	-0.131690	0.226744	0.079273	0.075958	-0.298119	-0.226584	-0.054159	-0.049139	0.132286	0.692089	0.325982	-0.093746	0.073123	-0.227742	-0.043880	-0.035310
Grad.Rate	0.252316	-0.169241	-0.208065	0.269129	-0.109268	0.216163	0.559944	-0.005336	0.041904	-0.590271	0.219839	0.122107	-0.069197	0.036477	-0.003394	-0.005008	-0.013071

Table 7. PCA datarame

2.7 Write down the explicit form of the first PC (in terms of the eigenvectors. Use values with two places of decimals only).

The explicit form of first PC is as follows :

PC1 = 0.25 * Apps + 0.21 * Accept + 0.18 * Enroll + 0.35 * Top10perc + 0.34 * Top25perc + 0.15 * F. undergrad + 0.03 * P. undergrad + 0.30 * Outstate + 0.25 * Room. Board + 0.06 * Books + (−0.04 * Personal) + 0.32 * PhD + 0.32 * Terminal + (−0.18 * S. F. ratio) + 0.21 * perc. alumni + 0.32 * Expend + 0.25 * grad. rate

2.8 Consider the cumulative values of the eigenvalues. How does it help you to decide on the optimum number of principal components? What do the eigenvectors indicate?

Cumulative eigenvalues are as follows :

[0.32020628, 0.58360843, 0.65261759, 0.71184748, 0.76673154, 0.81657854, 0.85216726, 0.88670347, 0.91787581, 0.94162773, 0.96004199, 0.9730024 , 0.98285994, 0.99131837, 0.99648962, 0.99864716, 1.]

Using the cumulative eigenvalues we can see that the first seven principal components only capture 85% of the data. Thus, we can prioritise our Principal components and select the first seven PCs for further data processing.

Eigenvectors (principal components) indicates the direction of the new feature space. The eigenvector is the direction of that line while the eigenvalue is a number that tells us how the data is spread out on the line which is an eigenvector.

2.9 Explain the business implication of using the Principal Component Analysis for this case study. How may PCs help in the further analysis?

PCA helps in dimensionality reduction, it converts set of correlated variables to non-correlated variables and helps in dealing with multi-collinearity before a model is developed. For further analysis **7 PCs** will be sufficient.

In the above case study PCA helps us in dimensionality reduction by reducing 17 PCs to 7 PCs as 85% of the data is captured by these seven PCs.

Also, PC1 has the highest variance ratio and after studying the coefficients of these 7 PCs we can conclude that :

- The most important feature in PC1 is top25perc as it has the highest coefficient.
- In PC2 F.undergrad is the most important feature.
- In PC3 books is the most important feature.
- In PC4 apps is the most important feature.
- In PC5 room.board is the most important feature.
- In PC6 books is the most important feature.
- In PC7 personal is the most important feature.